Accelerating cosmology

Handley Lab: KICC Welcome Day

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Accelerating cosmology from theory to data

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Big problems in cosmology

- **Dark energy & tensions**: cosmic acceleration, H_0, S_8 – new physics or systematics?
- **The next surveys**: LSST supernovae/quasars, Euclid/JWST lensing, galaxy simulations.
- **Fundamental gravity**: actions → cosmological data; real-time GWs → Einstein Telescope.

Shared tools: JAX / BlackJAX + agents

- **GPUs**: classical Bayes / nested sampling.
- **AI for scientists**: agents for code and algebra; humans choose the physics and test the results.



Already working

89 s: binary neutron-star inference
Median on 128 s signals. [2607.28265]
Yallup, Prathaban et al.

25,602 latent parameters: **11 min / GPU**
ALCS. [2603.26644]
Lovick, Yallup & Handley.

Postdocs & fellows: Natalie Hogg, Matt Grayling, Chris Lovell, James Alvey, Nikhil Sarin.
PhD students: Dily Ong, Namu Kroupa, Sam Leeney, Toby Lovick, Lawrence Berry, Charlotte Priestley, Farah Wallauer, Selma Vangstein, Sam Hewson.